

Closed-Loop Dynamic Biasing for Threshold Voltage Stability in LEO Beamforming Systems

Kaden Du and James Jagielski



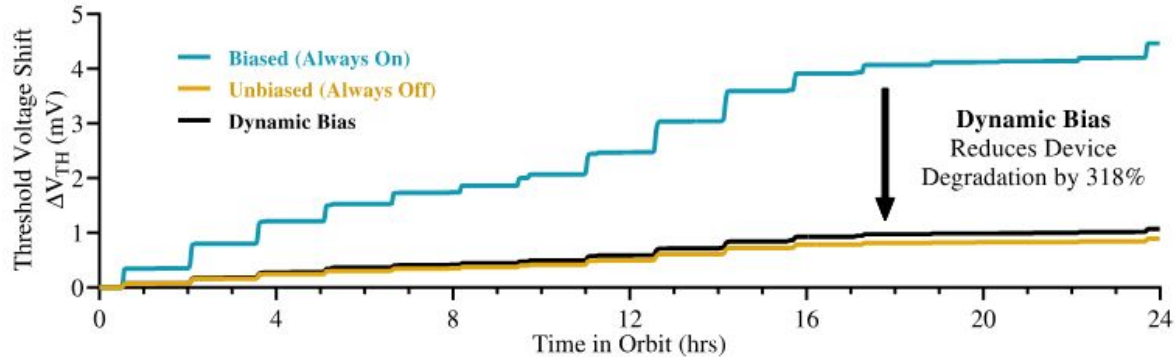
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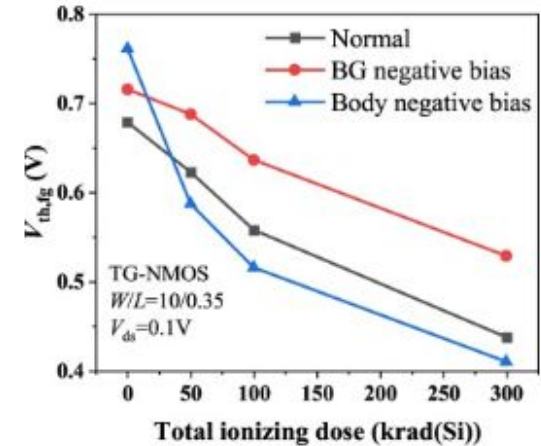
Introduction

- Engineering a system that performs consistently in the harsh environment of LEO is difficult, as threshold voltage shifts can induce changes in phase shift, oscillation frequency, and transmit power of the required microwave transmitter.
- A dynamic biasing system that adjusts the transistor's bulk terminal can actively correct threshold voltage shifts, ensuring performance over a longer life time. Our project will specifically cover dynamic biasing for space based solar power systems.



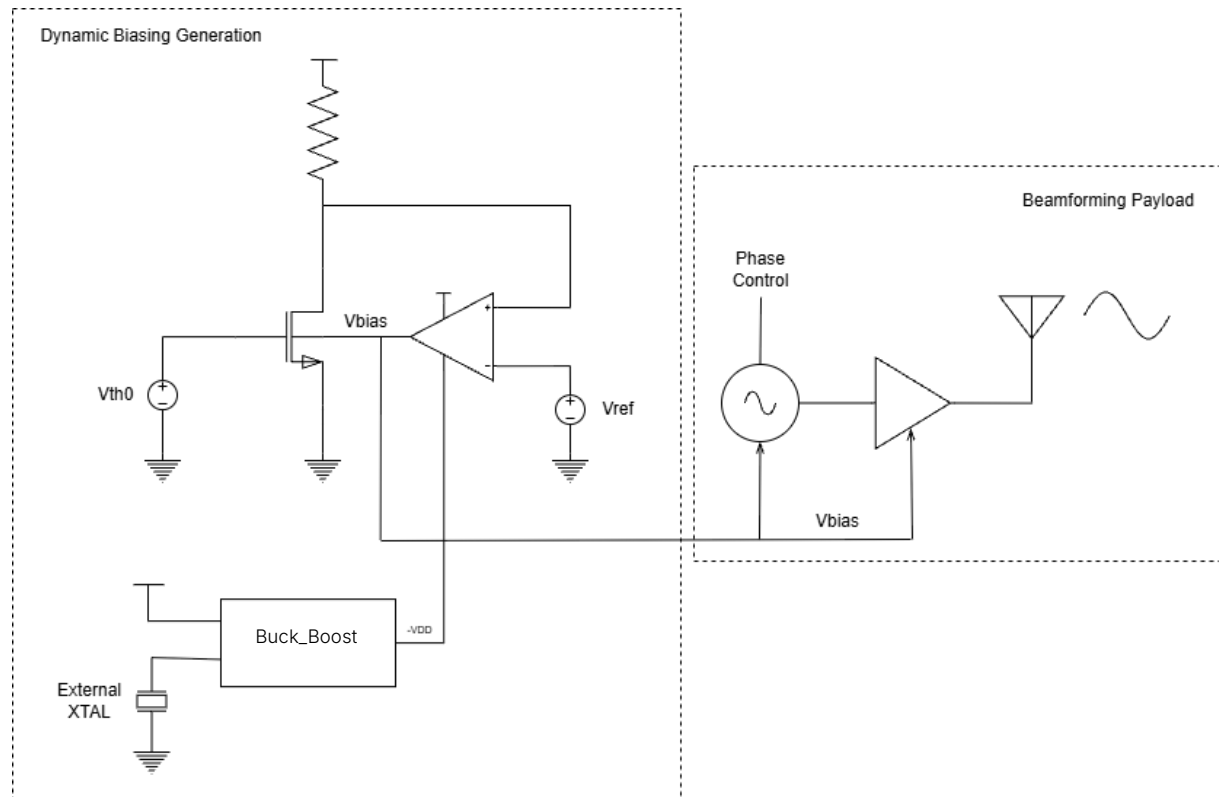
Background

- Back gate dynamic biasing has been previously shown to improve radiation hardness of MOS devices, namely from 50krads to 100krads according to one source [6]
- Typically, these designs are implemented using a TID sensing circuit, which drives the body voltage [2][6]
- These results do improve radiation hardness, however, any inaccuracy between the relationship between V_b and the TID V_{th} shift will show at high radiation doses
- As well, these designs can only account for the V_{th} shift due to radiation or temperature, not both
- Our project implements the dynamic bias through threshold voltage detection rather than TID or temperature sensing, defining a closed feedback loop for constant threshold voltage control.



Threshold Voltage shift over TID [6]

Design Overview

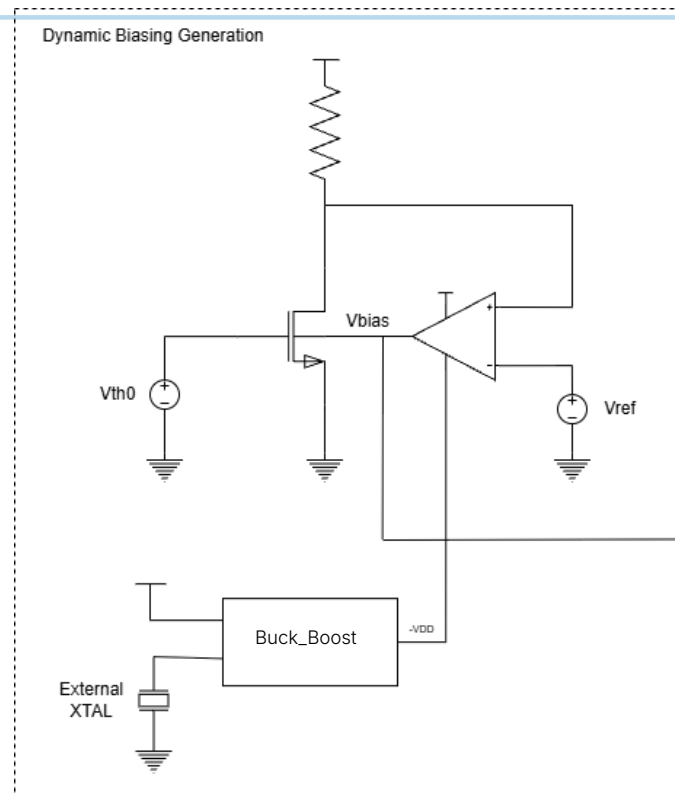


- Design broken down into the dynamic bias generation, followed by the application circuit.
- Design implemented in TSMC 65nm PDK
- All results from simulation in Cadence Virtuoso
- Simulating results based on temperature shift from -80C to 160C.



Dynamic Biasing Subcircuit

- Closed loop threshold voltage control
 - Feedback Amplifier drives bulk of all nmos transistors such that the desired threshold voltage is constant
 - V_{ref} and R set the current through the transistor to $10\mu A$
- Negative supply for dynamic biasing is generated by local buck-boost
- Body reference voltage is connected to all on chip NMOS transistors to stabilize the chip
- This design also gives control of threshold voltage for any application



Amplifier Design

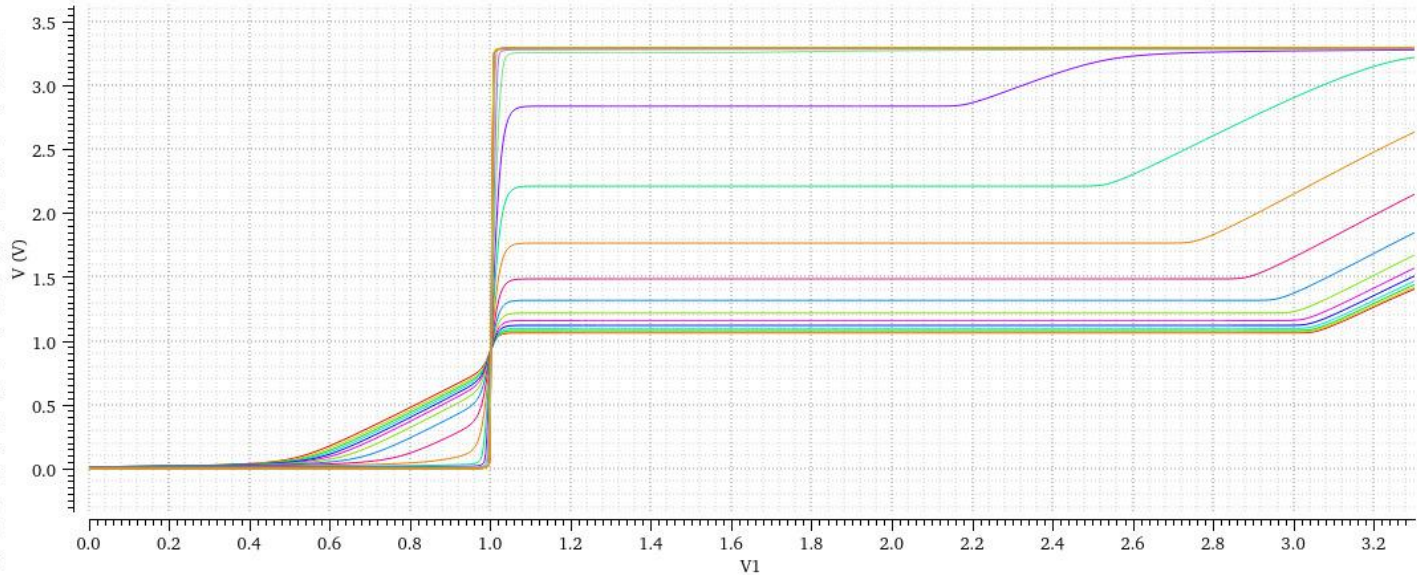
Here is a 3.3V op amp tested for VTC curves for a range of temperature from -200C to -150C

DC Response

Sun Dec 14 15:03:39 2025 1

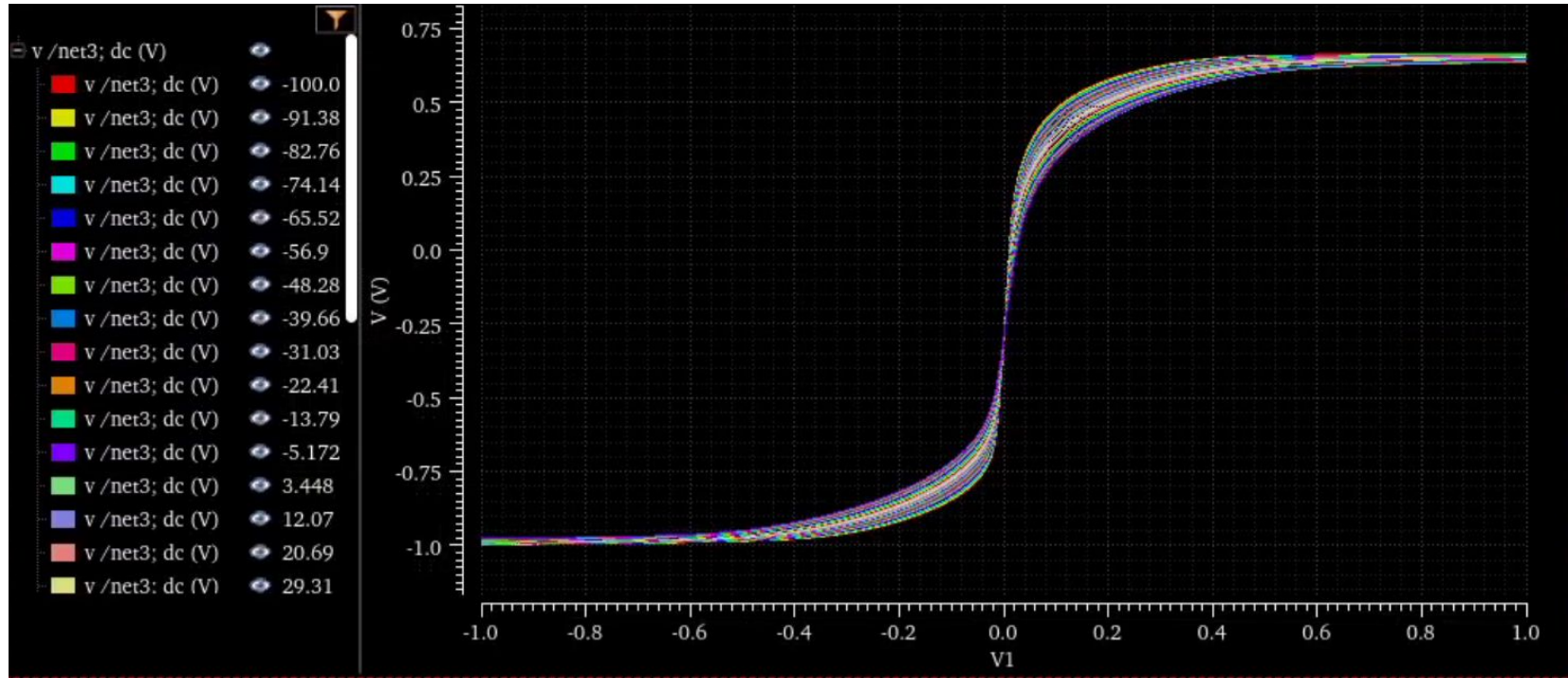
Name temp

/Vout		
■	/Vout	-200.0
■	/Vout	-197.4
■	/Vout	-194.7
■	/Vout	-192.1
■	/Vout	-189.5
■	/Vout	-186.8
■	/Vout	-184.2
■	/Vout	-181.6
■	/Vout	-178.9
■	/Vout	-176.3
■	/Vout	-173.7
■	/Vout	-171.1
■	/Vout	-168.4
■	/Vout	-165.8
■	/Vout	-163.2
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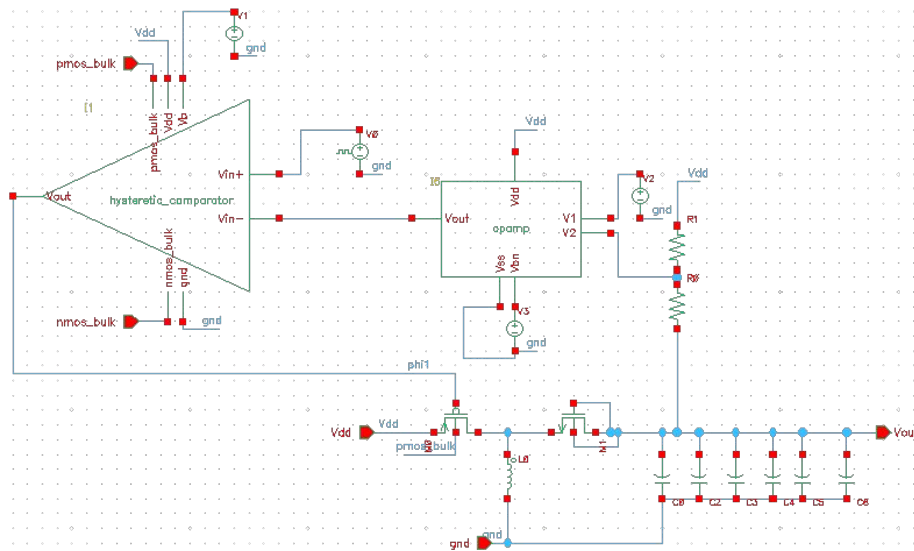
Amplifier Design

Here is a 1V opamp tested for VTC curves for a range of temperature from -100C to 150C

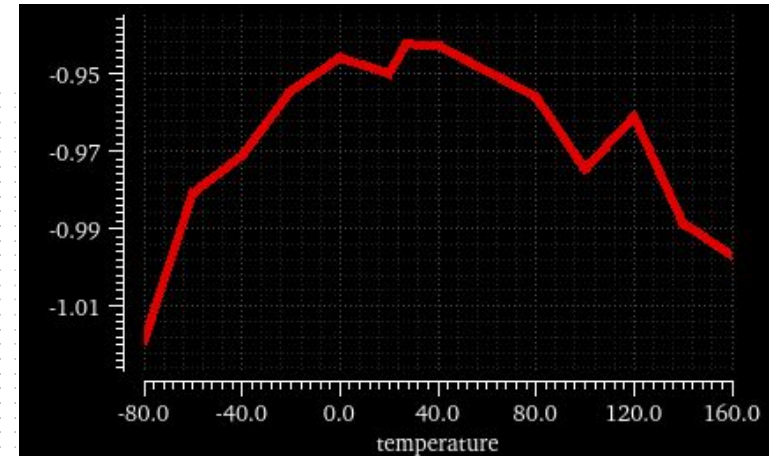


Buck-Boost Design

- Buck Boost Converter to supply -1 V to dynamic bias circuit
- Feedback and control circuitry can be stabilized over temperature using the dynamic biasing circuit
- Load is low and DC

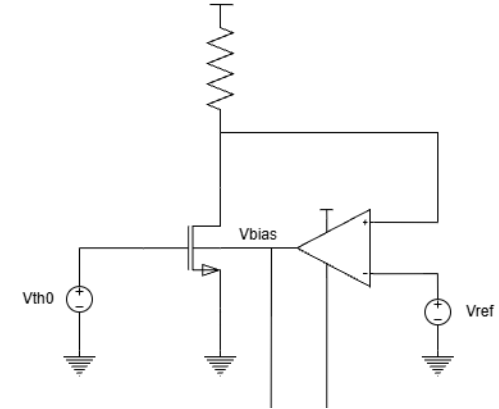
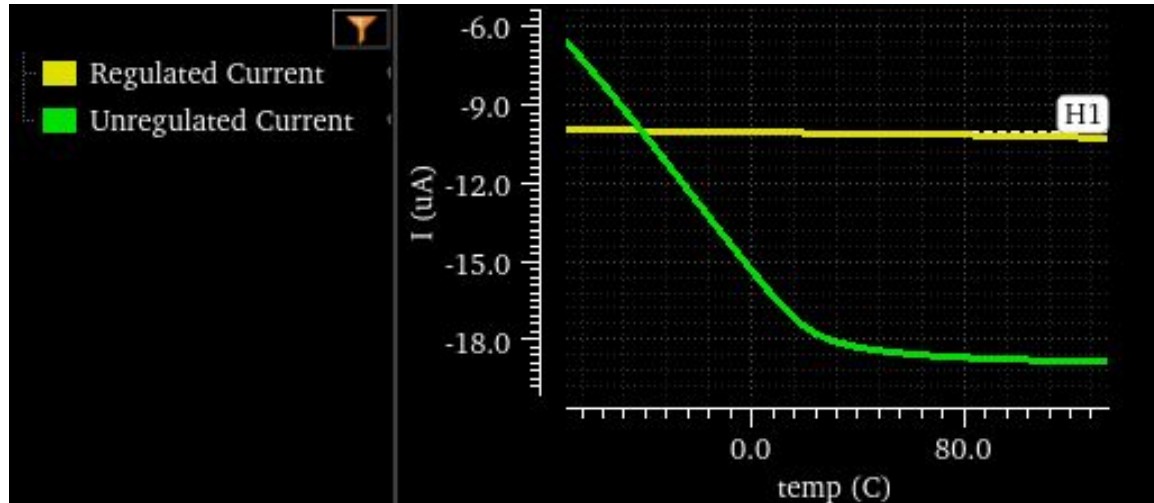


Regulated Voltage over temperature



- Results vary between -940mV to -1.02 V over temperature range, all within range of spec

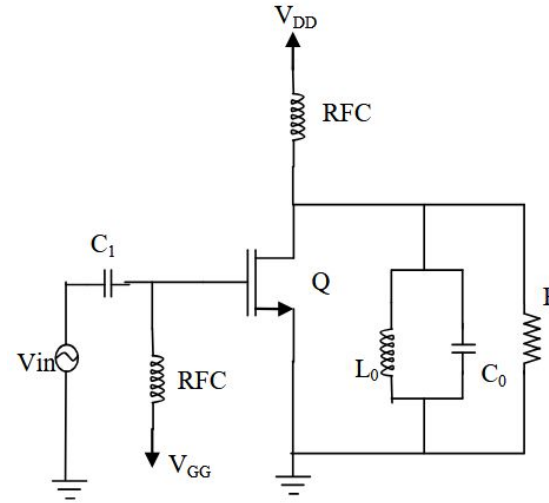
Dynamic Biasing Results



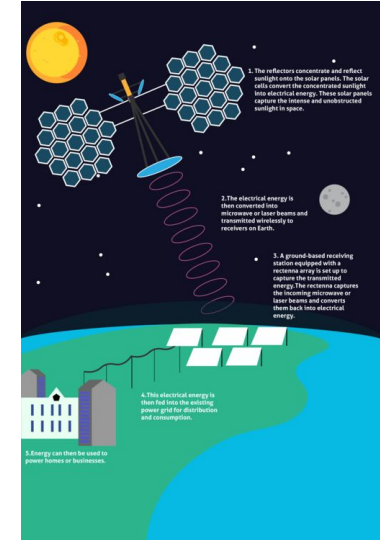
- Design successfully stabilizes drain current of transistor to the desired value, therefore stabilizing the threshold voltage

Application - Power Amplifier for Beamforming

- **Dynamic biasing is applicable to any system with radiation or temperature varying threshold voltage.**
- One such of these applications is Space Based Solar Power, a concept proposed to leverage
- LEO satellites, like the ones that would be used for SBSP experience:
 - 19Gy of TID over 1100 days [4]
 - -68 to 134 degrees Celsius [5], depending on levels of shielding
- To test dynamic biasing with this application, we will be simulating a microwave power amplifier with and without dynamic biasing to test performance differences.



Class C Power Amplifier Implementation
Experimentally validated by measuring output power



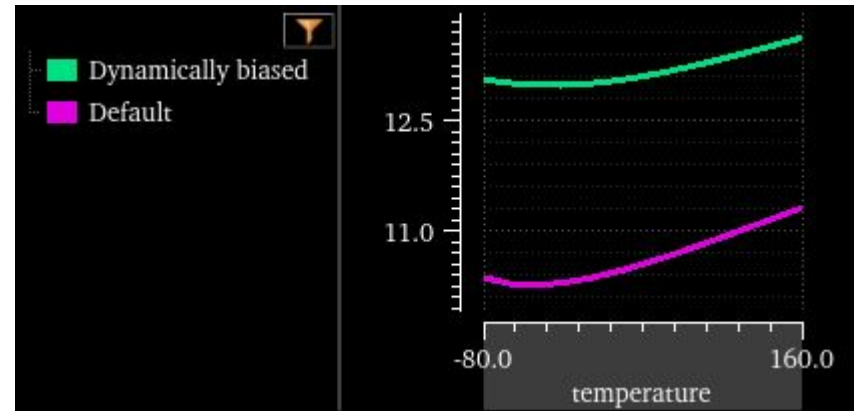
Application Results

- Output Gain:
 - Varies from 10-11.3dB in default case over temperature
 - Varies from 13dB - 13.6dB in dynamically biased case
- Higher gain due to artificial lowering of V_{th} in dynamic bias circuit, this serves as a 2dB gain benefit in this circuit
- Improved stability over temperature range, leakage current and higher mobility causing non perfect results

Continued tests

- Device should be taped out and tested in radiated environments
- Output power of power amplifier can be measured at multiple radiation steps, compared to nominal
- Output current of bias transistors can be measured as well

Dynamically Bias and reference output power for PA over temperature



Risks, Limitations & Mitigation Strategies

- Our design is using standard CMOS devices, which pose a problem because silicon is susceptible to temperature fluctuations and radiation effects.
 - A true space ready design would be packaged in GaN or SOI process which are known for space environment electronics.
- We have not done any longevity tests, which is vital for real life applications.
- The main feedback op amp for the bulk biasing could pose a threat
 - Our simulations show the reliability of the op amp in our temperature range, but there could be better topologies to use.
 - **Chopper-stabilized and auto-zero (AZ) techniques**
 - Has higher temperature range
 - Better response of radiation damage
 - Adjusts for offset for mobility, leakage, and noise
 - Longer lasting
- Bias range is limited as to not conduct through body diode
 - We can only raise the threshold by a set amount as and further bias would conduct through the PN junction
 - We can adjust the desired V_{th} to be lower as to give more headroom to adjust later on
- This technique does not compensate for Single Event Effects in space, and would need to be at least partially shielded, or implemented on a wide-bandgap semiconductor to mitigate these effects.



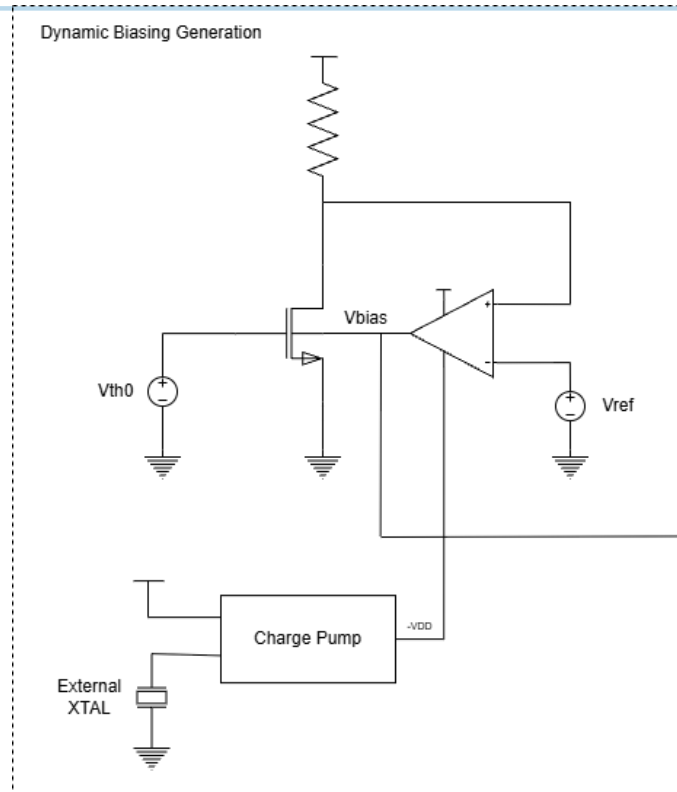
Future Work

- One limitation of this work is that it does not account for all changes due to temperature and radiation, such as leakage. While performance is improved, it is not made completely consistent.
- A common issue within harsh temperature environments is matched currents. We believe expanding our research into matching current sources would create more robust designs combatting the temperatures. We are also interested in expanding the temperatures to even colder and even warmer temperatures. This means in deep and space and even planetary missions could utilize these circuit techniques.
- Another route we would like to pursue is implementing this bias circuit onto GaN to ensure it works for newer space electronic technologies.

Conclusion

- In LEO satellites, as well as other systems, threshold voltage shifts due to TID radiation and temperature changes, degrading device performance
- Closed loop dynamic biasing has proved to stabilize current/threshold voltage over shifts due to temperature in simulation, and theoretically will correct
- Demonstrated to aid performance in the application of microwave power amplifiers for space-based solar power
- Next Steps
 - Test this technology in wide bandgap processes
 - Improve amplifier reliability across wider temperature/radiation range

Questions?



References

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